

Madhavan A Ramanujan

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SUMMARY

Computer Science undergraduate specializing in AI & Robotics with hands-on experience building deep learning and computer vision systems. Skilled in developing real-time AI applications using TensorFlow, Keras, OpenCV, and Flask, with strong foundations in Data Structures, Algorithms, Operating Systems, DBMS, and Computer Networks. Experienced in CNN-based models, transfer learning pipelines, and intelligent automation systems, with strong interest in AI-driven automation and software engineering.

EDUCATION

VIT Chennai — B.Tech Computer Science & Engineering (AI & Robotics) | 2023–2027 | CGPA: 8.08

TECHNICAL SKILLS

Languages: Java, Python, C, C++

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

ML: TensorFlow, Keras, CNNs, Transfer Learning, OpenCV, Computer Vision, OCR

Backend & Web: Flask, REST APIs, HTML/CSS

Tools: Git, GitHub, Jupyter Notebook

CERTIFICATIONS

- Oracle Cloud Infrastructure AI Foundations Associate
- Oracle Cloud Infrastructure Generative AI Professional

PROJECTS

AI-Powered Dyslexia Detection System (Python, TensorFlow, Flask, OCR, OpenCV) |

<https://github.com/Maddy152006/dyslexia-detection-ai>

- Architected a CNN-based character classification pipeline trained on 100,000+ handwritten glyph samples to detect dyslexic writing patterns across Normal, Reversal, and Corrected classes with ~96–97% validation accuracy
- Built a Flask-based OCR inference system integrating PyTesseract, OpenCV preprocessing, adaptive thresholding, and character segmentation for real-time dyslexia screening and annotation
- Implemented dataset augmentation, class rebalancing, confidence-based prediction logic, and confusion matrix evaluation using TensorFlow/Keras for explainable character-level analysis

Vision-Guided Robotic Waste Segregation System (Python, TensorFlow, OpenCV, MobileNetV2, Arduino Uno)

| <https://github.com/Maddy152006/waste-segregation-ai>

- Developed a transfer learning-based waste classification pipeline using MobileNetV2 to perform multi-class categorization across Metal, Paper/Cardboard, Plastic, and Trash classes with ~88% validation accuracy
- Built a real-time computer vision inference system integrating OpenCV, DroidCam webcam streaming, frame preprocessing, and live prediction overlays for automated waste detection and classification
- Integrated TensorFlow-based prediction outputs with an Arduino Uno-controlled robotic arm using serial communication, enabling automated servo-based waste segregation through calibrated angle positioning. Implemented multi-servo robotic arm control logic for category-specific movement automation and intelligent object routing within a real-time AI-assisted recycling workflow
- Applied dataset augmentation, transfer learning fine-tuning, and confusion matrix evaluation pipelines using TensorFlow/Keras to improve classification generalization for smart recycling automation systems

Algorithm Visualization Suite (Java, Java Swing, Data Structures & Algorithms) |

<https://github.com/Maddy152006/Algorithm-Visualizer-mini-project/tree/main/SRC>

- Engineered an interactive algorithm visualization system using Java Swing to simulate sorting and graph traversal algorithms with real-time animated execution flows
- Implemented Bubble, Selection, Insertion, Merge, and Quick Sort visualizations with dynamic state rendering, element highlighting, and event-driven UI controls
- Developed BFS and DFS graph traversal simulators using queue/stack-based execution logic, cyclic graph support, and real-time node traversal visualization for DSA learning and debugging

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **Competitive Badminton Player** – Won multiple junior badminton tournaments and competed in **Karnataka State Open and state ranking** badminton tournaments, demonstrating discipline, consistency, and competitive spirit.
- **Member, VIT Chennai Sports Club** (2024–2025).
- Participated in the **VIT Chennai 5K Run**.